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Mapping Election Officials' response to a high-misinfo & low trust environment during the 2024 U.S. election

MDI Scholar: Jorge Bris Moreno, MS in Data Science and Analytics, 2025
Research Team: Dr. Thessalia Merivaki, Dr. Mara Suttmann-Lea, Matt Steinberg, Priyasha Chakravarti, Kersyn McBride, Adi Vishahan, Santiago Vidal

Abstract

This study explores methods for thematically labeling social media content, balancing automated approaches and human coding to capture nuances in digital communications. Using data from the 2022 and 2024 Election Officials Communication Tracker, which categorizes over 60,000 posts from 1,500 state and local election officials' accounts on multiple platforms, we analyze communication strategies such as misinformation countermeasures, trust-building communications, and broader voter outreach. We employ a hierarchical taxonomy of 120 labels and compare automated labeling methods against rigorously validated human-coded data. The study focuses on assessing the relative accuracy of automated approaches, aiming to improve labeling processes and contribute to research accessibility in this field.

Introduction

The Election Officials Communications Tracker aims to document how election officials (EOs) communicate with voters about "*the times, places, and manner of holding elections*" in the United States. It involves the systematic data collection and analysis of state and local election officials' communications on mainstream social media platforms - Facebook, Twitter/X, Instagram, Threads, TikTok, and YouTube.

With over 120 thematic labels that capture communications about the election process, we aim to study Election Officials' voter education strategies and identify effective communications and gaps. The data and taxonomy of labels will be available for other researchers to use.

Labelling of social media content is still ongoing by our research group. To ensure satisfactory inter-coder reliability, coders engaged in 4 rounds of coding exercises. Considering that this is time consuming, the purpose of this project is to reduce the time spent in human labelling. We are therefore exploring different algorithms to automate this process.

Data Collection

The data is collected through Junkipedia, a social listening platform that enables users to identify, track, and analyze civic discourse across the internet. Our core dataset includes posts shared by U.S. Election Officials' official accounts during the 2024 election. The dataset includes post metadata, multimedia (images/videos), embedded text or audio translations, and the body text.

Figure 1 & 2 — Social Media Posts Examples
Easy (voter registration) and Hard labels (trust-building).

Labeling techniques

We are exploring three supervised learning model types: classic ML models, pretrained NLP models, and custom deep learning architectures. We are also experimenting with various embeddings/vectorizers, including TF-IDF, mixedbred embeddings, and Byte-Pair Encoding. We test these models on seven labels (Trust building, Election ready, GOTV, In-person, Mail-in, Media, Voter Registration)

To refine our methodology, we have categorized the labels into two groups: ***hard and easy labels***. Hard labels require more context and are more nuanced (i.e trust-building), while easy labels focus on key concepts and keywords (i.e. voter registration). This distinction enables us to optimize model selection and performance.

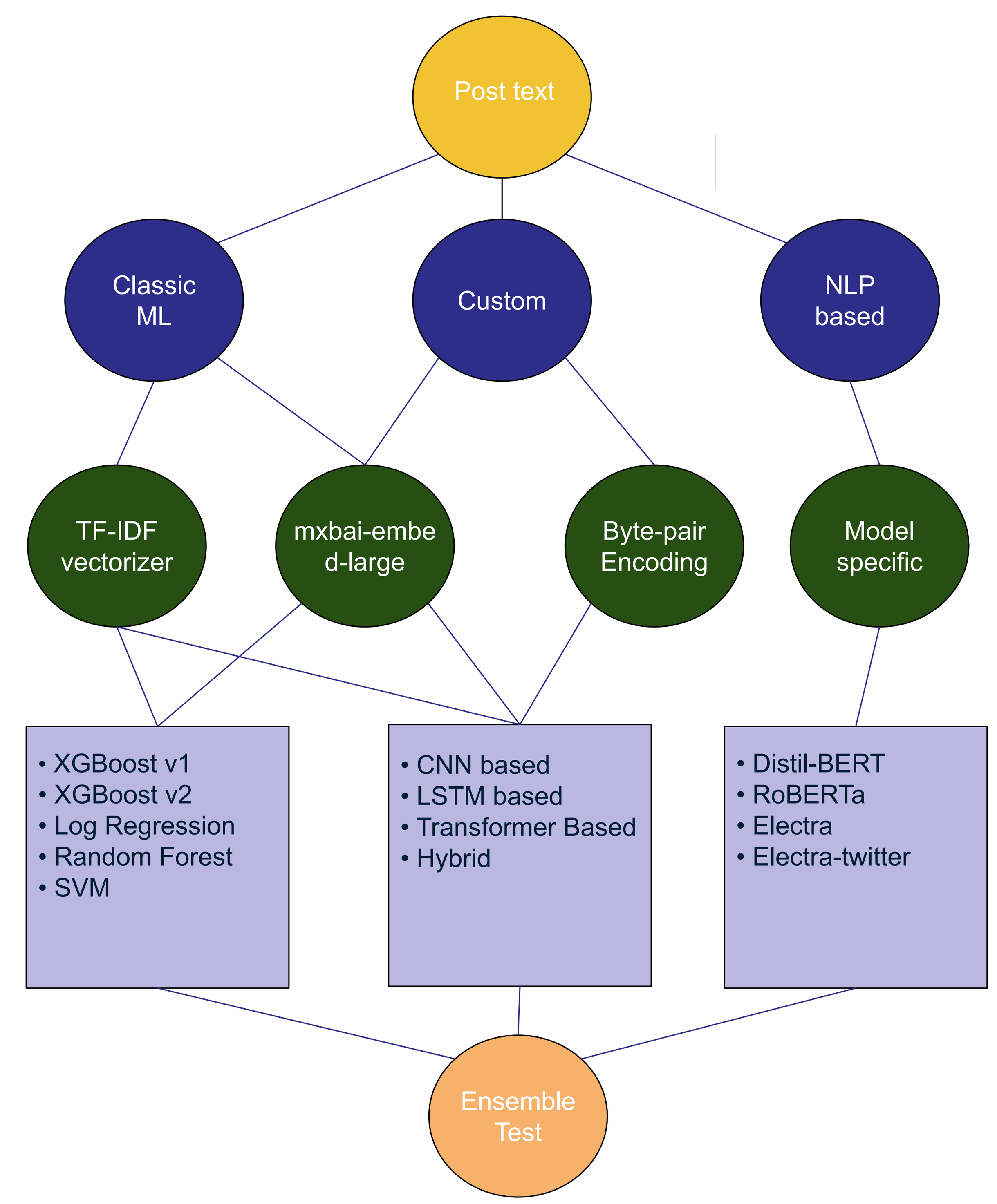

Figure 3 — Approach
The figure above represents the current approach for the labeling automation. Testing different combinations of models and vectorizers, we are assessing which models perform the best and if any are shared across different labels.

More: Tone Analysis (Biases)

With access to this dataset, we are also analyzing the "tone" of election officials' communications. We explore the presence of bias based on party affiliation, regional variations in communication strategies, and identifying potential missing labels through clustering. Techniques such as POS tagging and clustering are employed to gain deeper insights.

Ethical Considerations

Accuracy, representativeness, and reliability our research process and data output is of utmost importance.
Our dataset is heavily imbalanced, and the labels significantly impact the recommendations and analysis of voter information. This raises critical questions about which labels should be automated and whether to prioritize false positives or false negatives. Additionally, as communication strategies evolve over time, a more dynamic approach should be taken for training each model type, including the selection of vectorizers.

Conclusion / Future Directions

While we continue training models and identifying the best ones for seven of our labels, the next step is to train the top five models for the remaining labels we aim to automate. We are also evaluating high priority labels for our use case and considering whether to drop or broaden some, as working with 120 labels complicates our focus. Additionally, we will continue exploring alternative methodologies to streamline this automation process.

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